I. Introduction

We wanted to find the concentration of calcium and magnesium in commercial mineral water using EDTA. These two elements are typically found as polyatomic cations in natural water. Since they both form a 1 to 1 complex with EDTA the number of moles of EDTA that reacted with the solution will equal the moles of metal that is present in our conditions.[1]

There are two titrations that can be done to analyze mineral water. One is done at pH 10 where we can titrate our sample of buffer mineral water with EDTA to get a value for our Calcium and Magnesium together. Using Eriochrome Black T the endpoint should appear as a color change from wine red to blue. We will titrate another sample of our mineral water after saturating it with sodium hydroxide which will cause the magnesium to precipitate out of the solution as Mg(OH)2. We will use Hydroxynaphthol Blue to find our value of Calcium. Then take the molarity of calcium and subtract it from our total molarity to get magnesium.

The equations below depict what is happening. Y4- is EDTA in its fully dissociated form. Depending on pH there will be more or less protons bound to EDTA but these equations just show how they react with each other.

$$Ca^{2+} + Y^{4-} \rightarrow CaY^{2-}$$
$$Mg^{2+} + Y^{4-} \rightarrow MgY^{2-}$$
$$Mg^{2+} + 2OH^{-} \rightarrow Mg(OH)_2$$

II. Data and Analysis

Total metal measurement	Initial	R1	R2	R3	Blank
Mineral water	25	25	25	25	0
Nanopure water	25	25	25	25	50
pH 10 buffer	3	3	3	3	3
Initial burette reading	4	19	32.8	0.6	14
Final burette reading	17	32.8	46	13	18
EDTA used in analysis	13	13.8	13.2	12.4	4
Blank corrected EDTA	9	9.8	9.2	8.4	—

The final burette readings were taken when the solution in the receiving vessel got a color change, but they were not exactly the same color, which is a source of human error.

alcium measurement	Practice	S2	S3	Blank
Mineral water	25	25	25	0
Nanopure water	25	25	25	25
Initial burette reading	—	36	5.5	14.8
Final burette reading	—	46.3	14.8	14.8†
Recorded EDTA delivered	—	10.3	9.3	0
Blank corrected EDTA	—	10.3	9.3	—

The Practice trial is just to roughly estimate the calibration point so no exact measurements were taken.

III. Calculations

- EDTA concentration

n(EDTA) = mass/ molar mass
C(EDTA) = n(EDTA) / final solution volume
n(EDTA) = 0.3137 g/ 372.24 g mol^{-1} = 8.42736 × 10^{-4} mol
C(EDTA) = 8.42736 × 10^{-4} mol ÷ 0.250 L= 0.003371 mol L^{-1}.

- EDTA volume and blank correction

V(delivered) = V(final) − V(initial)
R2: 46.0 mL − 32.8 mL = 13.2 mL.
V(corrected) = V(delivered) − V(blank)
R2: 13.2 mL − 4.0 mL = 9.2 mL.

Trial	Delivered EDTA (mL)	Blank (mL)	Corrected EDTA (mL)
Initial	13	4	9
R1	13.8	4	9.8
R2	13.2	4	9.2
R3	12.4	4	8.4
Calcium S2	10.3	0	10.3
Calcium S3	9.3	0	9.3

- Total calcium plus magnesium concentration

C(total) = C(EDTA) × V(corrected)/ V(sample)
R1: C(total) = (0.003370943 mol L^{-1} × 0.0098 L) / 0.02500 L\
R1: 0.001321410 mol L^{-1} = 1.321 mmol L^{-1}.

Initial (mmol/L)	R1 (mmol/L)	R2 (mmol/L)	R3 (mmol/L)
1.214	1.321	1.241	1.133

- Averages and magnesium by difference

Mean corrected total volume = (9.0 + 9.8 + 9.2 + 8.4) mL ÷ 4 = 9.10 mL.
Mean C(total) = (0.003370943 mol L^{-1} × 0.00910 L) ÷ 0.02500 L = 0.001227023 mol L^{-1}.
Mean calcium volume = (10.3 + 9.3) mL ÷ 2 = 9.80 mL.
Mean C(Ca) = (0.003370943 mol L^{-1} × 0.00980 L) ÷ 0.02500 L = 0.001321410 mol L^{-1}.
Mean Ca = (0.001321410 mol L^{-1})(40.078 g mol^{-1})(1000 mg g^{-1}) = 53.0 ppm.
C(Mg) = mean C(total) − mean C(Ca)
C(Mg) = 0.001227023 mol L^{-1} − 0.001321410 mol L^{-1} = −9.43864 × 10^{-5} mol L^{-1}.

Mg (mg L^{-1}) = C(Mg) × 24.305 g mol^{-1} × 1000 mg g^{-1}
Apparent Mg = (−9.43864 × 10^{-5} mol L^{-1})(24.305 g mol^{-1})(1000 mg g^{-1}) = −2.29 ppm.

This negative result is physically invalid.

- Relative standard deviations

Sample standard deviation: $s = [\Sigma(x_i - \bar{x})^2 \div (n - 1)]^{½}$
Relative standard deviation: RSD (%) = $(s \div \bar{x}) \times 100$
Total-metal corrected volumes:
$s = [(0.01 + 0.49 + 0.01 + 0.49)\ mL^2 \div 3]^{½} = 0.5774$ mL
RSD = (0.5774 mL ÷ 9.10 mL) × 100 = 6.34%

- Calcium corrected volumes:

$s = [((10.3 - 9.8)^2 + (9.3 - 9.8)^2)\ mL^2 \div 1]^{½} = 0.7071$ mL
RSD = (0.7071 mL ÷ 9.80 mL) × 100 = 7.22%

Within each series, concentration is proportional to corrected EDTA volume; therefore the concentration therefore the concentration and corrected-volume RSDs are equal.

Quantity	Mean result	RSD (%)
Total Ca + Mg	1.23 mmol/L	6.34
Ca	53.0 ppm	7.22
Mg	No valid result	Not reportable

IV Results and Discussion

By taking the total metals and using our 4.0 mL blank we got an average concentration of 1.23 mmol/L for the total concentration of metals and our concentration of Calcium was 1.32 mmol/L, which translates to roughly 53 mg/L. Since we have more calcium than there can be if it was calcium + magnesium our concentration of magnesium is going to be negative. This means we have some type of issue with our values that is not being accounted for. It should not be changed to a positive number.

The total metal blank plays a big factor in this calculation. As we saw above the 4.0 mL for our total metal blank is about 30% of our uncorrected average of 13.1 mL. Our high blank is reducing our total metal concentration but has no effect on our calcium concentration, which may account for the sign of our magnesium concentration. We could have been contaminated in the sample, went over the endpoint or even just wrote the wrong number down. If we were to change our blank to something more favorable it would be incorrect to do so. Using just R1-R3 still gives us an average of 9.13 mL corrected which is less than our average for calcium of 9.80 mL.

With the recorded color of our starting solutions being blue/greenish blue there may be some issues with those as well. We can't conclude that we may have overshot our endpoint or if the indicator did not react correctly.

V. Conclusion

Calcium and Magnesium were determined in mineral water using EDTA in buffered solution and in strong basic conditions. 0.3137 g of disodium EDTA dihydrate when made up to 250ml as instructed in the manual will give a concentration of EDTA of 0.003371mol/L. From the results we have been able to calculate a preliminary value for Total metal content of 1.23mmol/L and Calcium of 1.32mmol/L which equates to around 53mg/L of Calcium which when subtracted from each other gives us a negative number for magnesium. We cannot therefore provide a confirmed value for Magnesium and the content of minerals within this mineral water. More titrations need to be carried out on both blank and sample.

Reference

1. EDTA Titration of Ca^{2+} and Mg^{2+} in Water. In Laboratory Manual; Experiment 4; unpublished course handout, n.d.; Procedure, steps 1–6.